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CONSERVATION ENTERPRISES
LEARNING GROUP BRIEF

Assessing the Core Assumption that Enterprise Benefits Promote Conservation Attitudes and Behaviors

Summary of Selected
Resources

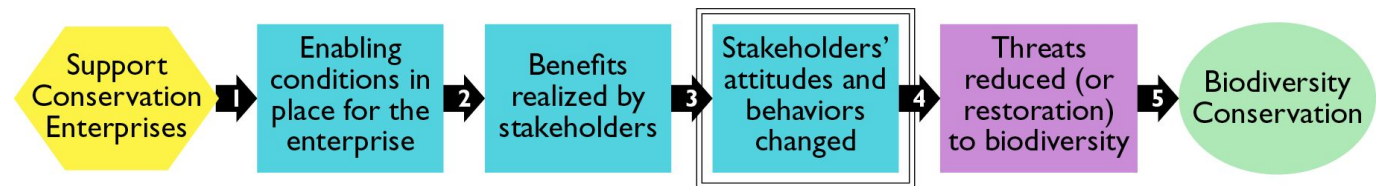
An achiote (*Bixa Orellana*).

Photo credit: Felipe Barney Arango/WWF-Colombia

Introduction

Contact the **Conservation Enterprises Learning Group** to share additional examples for inclusion in future resources.

One of the core assumptions in a theory of change for supporting conservation enterprises is that the benefits—both monetary and non-monetary—that participants receive promote desired attitudes and behaviors toward biodiversity conservation (see figure below).



Although about 25 percent of the U.S. Agency for International Development (USAID) biodiversity activities support conservation enterprises, there are few documented examples of assessments that examine the relationship between enterprise benefits and conservation behaviors—both within and beyond USAID. A systematic review of the literature by [Roe et al. \(2015\)](#) identified only a handful of studies providing such evidence.

INTRODUCTION Continued

During [a recent series of Conservation Enterprise Impact Labs](#), Conservation Enterprises Learning Group members discussed approaches to assessing assumptions in theories of change with USAID staff and their implementing partners. These discussions surfaced examples from other programs that test the core assumption that enterprise benefits promote conservation attitudes and behaviors. The following brief summarizes selected resources that assess how increased benefits have led to attitudes and behavior change. In taking a closer look at these studies, conservation enterprises teams should reflect on how they might also contribute to the evidence base for this key assumption.

Here you will find:

- [**Study:** Effectiveness of Gorilla Tourism and Other Livelihood Interventions around Bwindi and Mgahinga National Parks in Uganda](#)
- [**Impact Evaluation:** Three Economic Interventions to Reduce Deforestation in Protected Areas of Northern Cambodia](#)
- [**Study:** Impact of Sustainable Livelihoods on Cotton-Top Tamarin Conservation in Colombia](#)
- [**Quantitative Assessment:** Integrated Conservation and Development Project Involving Butterfly Farming in Tanzania](#)
- [**Assessment:** Loan Program for Linking Coastal Community Livelihoods to Marine Conservation in Aceh, Indonesia](#)
- [**Assessment:** Livelihood Benefits and Conservation Behaviors Resulting from Tourism-Conservation Enterprises in Kenya](#)
- [**Exploration:** Impact of Community Conservation Enterprises on Household Livelihoods Around Two Wildlife Areas in Sub-Saharan Africa](#)
- [**Resources**](#)

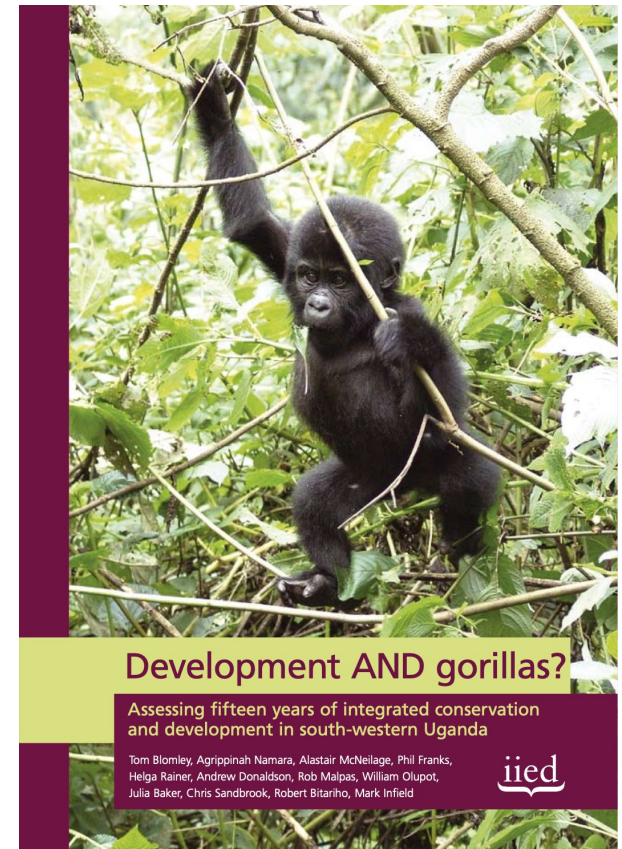
STUDY

Effectiveness of Gorilla Tourism and Other Livelihood Interventions around Bwindi and Mgahinga National Parks in Uganda

The **Institute for Tropical Forest Conservation, Bwindi and Mgahinga Conservation Trust, CARE International**, and the **International Gorilla Conservation Program** assessed the effectiveness of six integrated conservation and development interventions, including community-based gorilla tourism enterprises, implemented around the Bwindi and Mgahinga national parks in Uganda ([Blomley et al., 2010](#)).

METHODS

- Ranger survey.
- Community socioeconomic survey.
- Focus group discussions with community groups.
- Key informant interviews with local government representatives, resource users, and senior park staff.



STUDY Continued

KEY FINDINGS

- Community members benefiting from conservation and development interventions were more likely to have positive attitudes toward conservation than non-beneficiaries, even when disaggregated across all wealth categories.
- Engaging in tourism resulted in increased levels of cooperation between participants and park authorities, reporting of illegal activities, and willingness to assist in fire control.
- Diversifying income sources (through mushroom growing, honey production and processing, and handicraft making) reduced the need for poorer community members to harvest and sell forest products.

The study assessed if those benefiting from the interventions improved their attitudes toward conservation, cooperated with park management, and reduced illegal and unsustainable resource uses.

Three Economic Interventions to Reduce Deforestation in Protected Areas of Northern Cambodia

The **International Initiative for Impact Evaluation** assessed the effectiveness of the Wildlife Conservation Society's conservation interventions to improve household wellbeing and reduce agricultural deforestation in and around three protected areas in Northern Cambodia ([Clements, T. Neang, M, et al., 2020](#)).

METHODS

- A 2008-2017 monitoring program that regularly administered a socio-economic survey to [households](#) and [village chiefs](#) in 11 villages inside the protected areas and five villages outside them.
 - Evaluators used three measures of household wellbeing to assess the social impact:
 - ✓ The Basic Necessity Survey using a list of 35 items of household assets and basic services.
 - ✓ The total annual rice production.
 - ✓ A calculation of the total annual rice harvest and expected consumption to indicate food security.
- An assessment of farmers participating in IBIS Rice production and their behaviors related to converting forest to agricultural land.
- A randomized control trial set up by the evaluators to randomly allocate new participants into either the implementation group or the control group. Implementation group members could participate in IBIS Rice, while those in the control group needed to wait to do so.

IMPACT EVALUATION Continued

KEY FINDINGS

- Compared with the direct payments scheme and ecotourism opportunity, participation in the IBIS Rice intervention correlated more positively with increased economic status, increased rice harvest, and improved household food security between 2014 and 2017.
- There was an overall positive relationship between participation in production of IBIS Rice and a reduction in overall deforestation rates and forest land clearing.
- In areas of IBIS Rice production, agricultural deforestation decreased as much as 75 percent during the first year. The other two interventions had mixed results on reducing agricultural deforestation.



Photo credit: Sansom Mlup Prey

The Wildlife Conservation Society supported three livelihood interventions: a direct payments scheme to protect the nests of globally threatened birds; eco-tourism to support villagers in bird and habitat conservation; and IBIS Rice, a conservation enterprise that produces conservation friendly products sold at premium prices.

STUDY

Impact of Sustainable Livelihoods on Cotton-Top Tamarin Conservation in Colombia

Proyecto Tití is a multi-disciplinary conservation program in Colombia that supports sustainable development programs for communities near forests with cotton-top tamarins. A recent study ([Maynard et al., 2021](#)) evaluated whether sustainable livelihood activities like selling eco-mochilas or plush toys and engaging in plastic recycling or farmer conservation agreements made a positive difference in supporting livelihoods and contributed to changes in conservation attitudes and practices toward cotton-top tamarins.

METHODS

- Household surveys with participating communities, their neighbors, and members of reference communities nearby for comparison.
- The survey measured respondents' agreement, neutrality, or disagreement with statements about expected increase in income, improved wellbeing, sustainable resource use, and attitudes toward cotton-top tamarins.
- As Proyecto Tití lacked baseline data, this was a “post hoc” evaluation.



Photo credit: Mia Kennel

STUDY Continued

KEY FINDINGS

- Participants had higher (and more steady) income than their non-participating neighbors as well as greater access to resources for improved wellbeing (e.g., ability to access doctors, help family members, buy drinking water, and invest in new businesses).
- Participants did not increase the use of resources from the forest, a trend that conservation farmer agreements reinforced.
- Although participants, neighbors, and members of reference communities all agreed they should not keep cotton-top tamarins at home, participants did not always report others involved in the illegal tamarin trade. This finding that the illegal trade continues, and is not always reported by participants, suggests a need to focus both on business skills training for new participants in livelihood activities, and to re-focus on conservation awareness-building to reinforce conservation attitudes.

Integrated Conservation and Development Project Involving Butterfly Farming in Tanzania

The **Amani Butterfly Project** supports butterfly farming by community members in the East Usambara Mountains of Tanzania. This assessment sought to identify if there is a causal relationship between local butterfly farming income and forest conservation ([Morgan-Brown et al. 2010](#)).

METHODS

- Quasi-experimental study design that compared 150 butterfly farmers with a control group of 170 fellow community members using a survey of conservation attitudes, beliefs, knowledge, and behavior.
- Survey questions related to self-reported conservation behaviors in the past year, general attitudes toward local conservation, attitudes about effectiveness of conservation behaviors, and knowledge of conservation behaviors.

QUANTITATIVE ASSESSMENT Continued

KEY FINDINGS

- The amount and reliability of income from butterfly farming correlated positively with participation in forest conservation.
- Participation in forest conservation depended proportionally on the amount of income farmers earned. Those who earned less income from the enterprise were less involved in conservation efforts.
- Because butterfly farmers perceive a link between earnings from butterfly farming and conservation, participation in butterfly farming increased participation in conservation behaviors.



Hypolimnas Usambara or the Red Spot Diadem of Tanzania
Photo credit: Adobe Stock

Loan Program for Linking Coastal Community Livelihoods to Marine Conservation in Aceh, Indonesia

The **Iboih Makmur Cooperative Loan Project** included six livelihood borrowing groups: community-based tourism, fishing, trading, agriculture, livestock, and women. The study aimed to understand the performance of the project in reducing destructive fishing practices in the post-tsunami community of Iboih, Indonesia ([Novriyanto et al., 2012](#)). The authors assessed the demographic and socio-economic status of the loan recipients, potential impacts to long-term livelihood benefits, locally perceived livelihood benefits, and change in environmental attitudes and behavior.

METHODS

- A quasi-experimental approach in which researchers conducted semi-structured interviews with 103 of the 117 loan recipients, who they compared against a control group of 29 randomly selected community members.
- Interview questions addressed:
 - Livelihood constraints before and after the fund's establishment and key attributes leading to removal of those constraints.
 - Why people borrowed from the fund, changes in attitudes toward the fund (and why), perceived success of the fund, perceived linkages between conservation and the livelihood project, and changes in attitude and behavior toward the environment (and, where relevant, what had stimulated this).

ASSESSMENT Continued

KEY FINDINGS

- About 30 percent of the respondents positively changed their attitudes toward the environment since receiving the revolving fund loan. Out of the remaining respondents, about 40 percent already had positive attitudes and behaviors and about 30 percent had neither positive nor negative attitudes.
- Respondents who experienced a positive change in attitude toward the terrestrial and marine environment increased their awareness of the importance of protecting forests and the ocean (e.g., coral reefs), the negative impacts of degraded ecosystems (e.g., vulnerability to natural disasters), and the benefits of conservation (e.g., the linkages between tourism and ecosystem services and sustainable forest use and alternative livelihoods).
- At least 15 percent of participants changed their destructive fishing behaviors, including coral disturbance through dynamite fishing.
- Participants were more likely to confront offenders and report them to the customary marine leader.
- Findings were consistent with the Iboih Makmur Cooperative Loan Project's built-in environmental impact assessments and field checks by Fauna and Flora International. However, the authors cautioned that Fauna and Flora International's project activities during the same period, which aimed to strengthen local governance systems and empower customary marine leaders to reduce marine conservation threats, could have caused the aforementioned behavioral changes.

ASSESSMENT

Livelihood Benefits and Conservation Behaviors Resulting from Tourism-Conservation Enterprises in Kenya

A partnership between community land owners and a private investor established two tourism-conservation enterprises (TCEs): **The Sanctuary at Ol Lentille** (an eco-lodge) and the **Koija Starbeds** (a tented camp). **The African Wildlife Foundation** served as a third, neutral partner to promote a fair deal between the two parties, ensure donor funding, and monitor outcomes. This assessment examined the extent to which participation in the TCEs has helped improve community livelihoods and wildlife conservation ([Nthiga et al. 2015](#)).

METHODS

- A review of secondary data (literature review and project documentations).
- Semi-structured interviews and focus group discussions.
 - Interview participants came from three main stakeholder groups:
 - ✓ TCE group ranch leaders and members
 - ✓ The directors of the private investors
 - ✓ African Wildlife Foundation staff
 - The questionnaire addressed the establishment of the TCEs, decision-making processes, income generation and distribution, and governance challenges.
 - The authors used the five livelihood capital assets of the Sustainable Livelihoods Framework (human, physical, social, financial, and natural capitals) to analyze the TCEs' contribution to livelihoods. They assessed the TCEs' effectiveness in improving wildlife conservation by examining their impacts on biodiversity threats, including human-wildlife conflicts, land-use changes, land degradation, and poaching.

ASSESSMENT Continued

KEY FINDINGS

- Both TCEs resulted in significant contributions to participants' employment, education, healthcare, and security. The enterprises also resulted in additional leverages of indirect funding (from philanthropy, grants and donations, and guests staying at the lodges) for development projects.
- The TCEs had both positive and negative impacts toward biodiversity conservation.
 - Existence of TCEs ensured the communities' collaboration and partnership in managing the exclusive conservation areas.
 - Communities became more tolerant of wildlife (i.e., lack of retaliatory attacks or revenge on wildlife after livestock loss), partly because of the livelihood benefits from the eco-lodges and other related projects.
 - Negative impacts included incidences of poor commitment to sustaining soil erosion strategies, lack of respect for zoning rules, and illegal poaching. The increased population placed more pressure on conservation areas from illegal grazing and crop farming. Some enterprise beneficiaries reported using the increased income from the TCEs to purchase livestock.

Prior to the TCEs' establishment, the **African Wildlife Foundation** supported zoning designation (conservation, settlement, and grazing zones) and initiated soil erosion control initiatives. These efforts resulted in more than 7,050 hectares of exclusive conservation areas by 2011, leading to vegetation recovery that has helped conserve important wildlife habitats.

EXPLORATION

Impact of Community Conservation Enterprises on Household Livelihoods Around Two Wildlife Areas in Sub-Saharan Africa

The project included supporting the residents living adjacent to the **Greater Virunga Landscape in Rwanda and Uganda** with community conservation enterprises to provide benefits from tourism, including direct income, to improve household livelihoods (food, education, health, and financial securities). Due to land scarcity, high population pressure, and poor agricultural productivity, nearby communities had participated in illegal activities such as poaching and collecting resources illegally, which became a threat to park resources and management. This study evaluated impacts on the adjacent communities ([Sabuhoro et al. 2021](#)).

METHODS

- Face-to-face interviews with 571 randomly selected heads of households, with both closed and open-ended questions to assess satisfaction with livelihood security.
- Researchers used a stratified sampling frame to select interviewees from across the range of community conservation enterprises (i.e., craft-making cooperatives, eco-lodges, and cultural villages). Most participants were reformed poachers and illegal forest resources collectors.
- Adapting the [Sustainable Livelihood Framework](#) from Munanura et al., researchers asked interviewees to rate statements representing measures of the four aspects, on a 7-point scale from 1 (not satisfied) to 7 (completely satisfied) between a fixed point in time in the past (benchmark) and the current time of the interview.

EXPLORATION Continued

KEY FINDINGS

- The livelihood security of residents in Rwanda had significant improvements compared to those in Uganda, due to Rwanda's substantial tourism growth. The broader scope of tourism operations and revenue contributed to the growth of accommodation infrastructure, craft enterprises, and cultural tourism centers, creating more benefits for communities.
- The study highlighted the importance of integrating other enterprises, such as agriculture or beekeeping, in areas with low visitation numbers.



Children produce drawings inspired by Bwindi Impenetrable National Park for sale to tourists along the trails used for gorilla trekking tours and cultural walks. Buhoma, Uganda in September 2017.
Photo Credit: Jason Houston for USAID

With increased revenues from gorilla tourism, as well as other community conservation enterprises incorporating crafts centers, community lodges, and cultural villages, the project provided much-needed incentives to sustain wildlife conservation and enhance community livelihoods. To evaluate impact on the adjacent communities, the researchers assessed community members' perceptions regarding trends in their overall satisfaction with their household livelihood security, indicators of four dimensions of household livelihood security (food, education, health, and financial), and differences in satisfaction with livelihood security in comparison to the non-participant residents.

Resources

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- Sabuhoro E., et al. (2021). "Exploring the Impact of Community Conservation Enterprises on Household Livelihoods Around Two Wildlife Areas in Sub-Saharan Africa." *Human Dimensions of Wildlife*, DOI: [10.1080/10871209.2021.2015487](https://doi.org/10.1080/10871209.2021.2015487).



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To learn more about the Conservation Enterprises Learning Group visit:
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